changes in a collection of matter called system (a cell or a plant).

The energy exchanges between the system and the surrounding balance each other. All chemical reactions in cells involve energy transformation. (For example: Photosynthetic bacteria transform radiant energy into chemical energy). In living cells thermodynamic changes are essential for biological function such as growth, reproduction, photosynthesis and respiration. Microorganisms obtain energy and nutrients for their survival and reproduction through metabolism. The microbial species and ecological niche can often be differentiated from each other based on metabolic characteristics. The metabolic reaction often allows the use of micro organisms in fermentation process and biogeochemical cycle.



Three fourth of the energy is derived from carbohydrate that we consume and Glucose is the major fuel for all living organisms.

4.1 Metabolism

The term Metabolism refers to the sum of all bio chemical reactions that occur within a living cell. Chemical reaction either release energy or require energy. Metabolism can be viewed as an energy balancing act. It can be divided into two classes of chemical reactions namely **Catabolism** and **Anabolism**.

Catabolism: It is called catabolic or degradative reactions because complex organic compounds are broken down into simples ones. Catabolic reactions are generally hydrolytic reactions. It is enzyme regulated chemical reaction that release energy and they are exergonic Example: Break down of sugar into Carbon dioxide and water in cells.

Anabolism: It is called anabolic or biosynthetic reactions because complex organic molecules are formed from simples ones. Anabolic process often involves dehydration, are bio synthesis reactions (Figure 4.1). It is enzyme regulated energy requiring reaction and they are endergonic. Examples: Formation of proteins from amino acids.

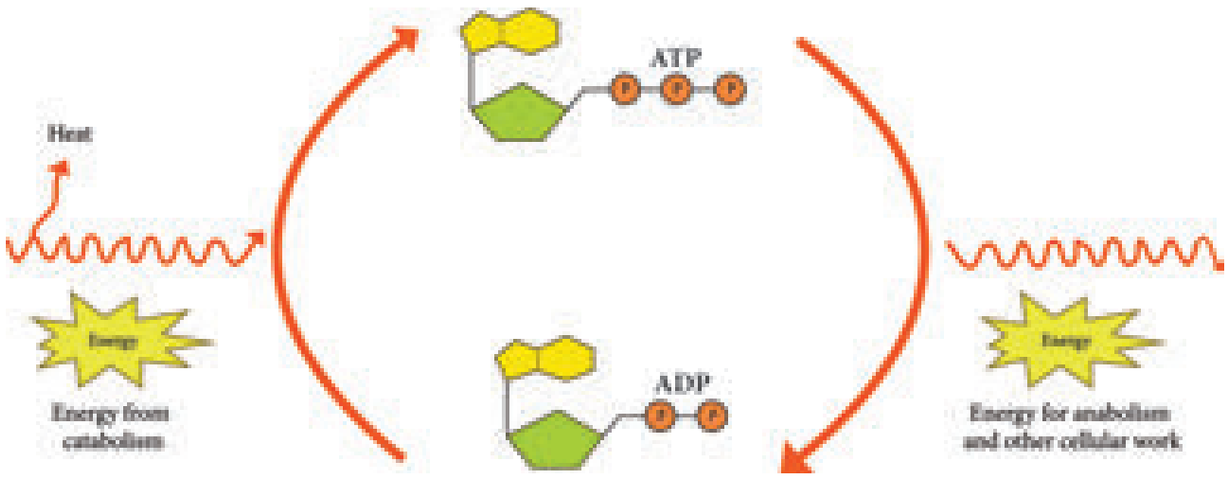


Figure 4.1: Catabolic and anabolic reactions

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